

AI-Based Fake Profile Detection on Social Media Using Three-Class Machine Learning Classification and Explainable Artificial Intelligence

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Abstract—The widespread proliferation of fabricated and bot-operated user profiles on social media platforms poses a growing threat to information integrity, public discourse, and digital trust. Conventional detection systems apply binary classification, producing a single legitimate-or-fake decision that is insufficient for real-world content moderation workflows where large numbers of accounts occupy an ambiguous behavioural grey zone. This paper presents a three-class detection framework that categorises social media profiles as *Real*, *Suspicious*, or *Fake* using an XGBoost gradient-boosted ensemble model trained on thirteen handcrafted features drawn from profile metadata, network topology, temporal activity patterns, and external bot-probability signals obtained from the Botometer Pro API. Training data were aggregated from four heterogeneous sources: the Twitter API v2, the GitHub REST API, the publicly available Cresci-2017 benchmark corpus, and programmatically generated synthetic profiles. Class imbalance was addressed through Synthetic Minority Over-sampling Technique (SMOTE) applied exclusively to the training partition to prevent data leakage. Evaluation on a held-out test set yielded an overall accuracy of 91.23%, a macro-averaged F1 score of 90.37%, and an AUC-ROC of 0.964. Model predictions are made interpretable through SHapley Additive exPlanations (SHAP), which identifies Botometer score, account age, and follower-to-following ratio as the three most discriminative features. The complete system is deployed via a FastAPI backend and a React.js web interface enabling real-time single-query analysis with per-class probability visualisation and feature importance charts.

Index Terms—fake profile detection, social media security, XGBoost, SMOTE, SHAP, explainable AI, bot detection, three-class classification, machine learning, Twitter

I. INTRODUCTION

Social media platforms have become foundational infrastructure for personal communication, political discourse, journalism, and commerce. Twitter alone reports over 600 million monthly active users, while Facebook and Instagram collectively host more than five billion accounts. This scale of participation has simultaneously made these platforms attractive targets for coordinated inauthentic behaviour, encompassing automated bot networks, impersonator profiles, and state-sponsored influence operations [1].

Fake or bot-operated profiles cause measurable harm across multiple domains. Disinformation campaigns amplified

through bot networks have demonstrably influenced electoral outcomes. Spam bots artificially inflate product ratings and distort search relevance. Phishing accounts impersonate financial institutions or public figures to defraud users. Coordinated bot swarms fabricate social consensus by generating artificial engagement with targeted content, distorting perceived norms in ways that can shift public opinion [2].

Despite the gravity of this threat, automated detection systems remain predominantly binary classifiers that assign a single label—either legitimate or bot. This binary framing is problematic for two reasons. First, a substantial proportion of accounts occupy a grey zone—possibly human-operated but exhibiting bot-like posting patterns, or newly registered accounts without sufficient behavioural history for confident classification. Coercing these accounts into a binary label introduces systematic misclassification. Second, binary labels do not support graduated content moderation responses: a platform may choose to immediately suspend high-confidence fake accounts, route suspicious accounts for human review, and take no action on confirmed genuine accounts [8].

This paper addresses these limitations through three primary contributions:

- A principled three-class taxonomy (Real, Suspicious, Fake) with a composite scoring mechanism that assigns labels based on a weighted combination of five independent signal categories.
- A multi-source labelled dataset aggregating live API data, curated benchmark corpora, and synthetic profiles totalling 11,247 profiles across all three classes.
- An XGBoost classifier with SMOTE-based class balancing, early stopping, five-fold cross-validation, and SHAP-based post-hoc explainability achieving state-of-the-art three-class performance.

The remainder of this paper is organised as follows. Section II reviews related work. Section III describes the proposed methodology. Section IV presents experimental results. Section V discusses limitations and future work. Section VI concludes.

II. RELATED WORK

A. Early Statistical and Rule-Based Approaches

Ferrara et al. [1] provided one of the earliest comprehensive surveys of social bots, characterising their behavioural signatures and surveying detection methodologies. Their taxonomy distinguished between simple bots, cyborgs (human-assisted bots), and fully automated accounts, motivating subsequent multi-label formulations. Varol et al. [2] introduced the Botometer service (previously BotOrNot), which combines over one thousand features across six categories—network, user, friend, temporal, content, and sentiment—and achieves approximately 86% accuracy in binary bot detection. Yang et al. [4] demonstrated that follower-following graph structure alone carries strong discriminative power for detecting fake accounts through a random forest model.

B. Deep Learning and Graph-Based Detection

Kudugunta and Ferrara [5] applied recurrent neural networks to tweet content and metadata jointly, reporting an AUC of 0.99 in binary detection tasks. Graph Neural Networks (GNNs) have been applied to the social interaction graph: Wen et al. [6] modelled user interactions as a heterogeneous graph and employed a graph attention network to propagate label information through friendship edges, achieving 94.2% accuracy on the TwiBot-20 benchmark. Heidari et al. [7] fine-tuned BERT embeddings on Twitter biographies and tweet sequences, outperforming traditional machine learning baselines by four to seven percentage points on binary classification tasks. However, the substantial computational cost of transformer-based models limits their suitability for real-time single-query applications.

C. Multi-Class and Gradual Classification

Work on finer-grained classification is comparatively sparse. Rauchfleisch and Kaiser [8] introduced a “cyborg” category for accounts blending human and automated behaviour, demonstrating that binary labels degrade classifier performance when applied to borderline accounts. Echeverria and Zhou [9] applied clustering techniques to Twitter bot collections and found that spambots, political bots, and social bots form distinguishable clusters in feature space, motivating multi-class treatment. Cresci et al. [10] released the Cresci-2017 dataset containing genuine accounts alongside three bot sub-types, and showed that ensemble tree models outperform neural approaches when feature sets are carefully engineered.

D. Explainability in Detection Systems

Beskow and Carley [11] argued that black-box classifiers are insufficient in adversarial settings where operators seek to understand which features to evade detection. LIME and SHAP have been applied to bot detection models to identify the most influential per-prediction features. The present contribution extends this line of work by integrating SHAP TreeExplainer natively into the inference pipeline, providing per-query feature attribution in real time.

The key differentiator of this work is the combination of a three-class labelling scheme, multi-source data fusion, gradient boosting with rigorous leakage prevention, and real-time SHAP explainability within a deployable full-stack web application.

III. PROPOSED METHODOLOGY

The proposed framework comprises five sequential stages: data collection, composite scoring and labelling, feature extraction, model training, and deployment. Fig. 1 illustrates the complete end-to-end system architecture.

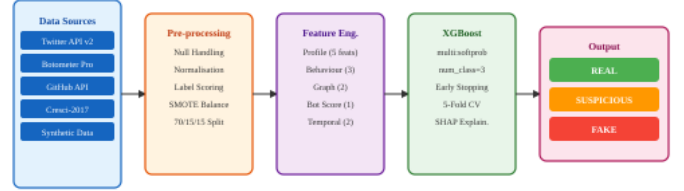


Fig. 1. End-to-end system architecture of the proposed detection framework showing data sources, preprocessing, feature engineering, XGBoost classifier, and three-class output.

A. Data Collection

Profiles were gathered from four complementary sources to maximise diversity across authenticity classes.

Twitter API v2 was queried using eight search terms strategically chosen to over-represent different user archetypes. Developer community queries (`#100DaysOfCode`, `#python`) were expected to yield predominantly genuine accounts, while spam-oriented queries (`bitcoin giveaway`, `followback`) were expected to yield predominantly fake or bot-operated accounts. For each Twitter user, the Botometer Pro API was invoked via its batch endpoint to obtain an independent bot probability score in the interval $[0, 1]$.

GitHub REST API was queried using six targeted search expressions differentiating highly active developers (real), recently created inactive profiles (suspicious), and accounts following hundreds of users with zero repositories (fake-like). GitHub profile metadata—repository count, follower and following counts, profile completeness, and account creation date—were mapped to the common feature schema.

Cresci-2017 [10] is a publicly available benchmark corpus containing 3,474 genuine accounts and 14,383 bot accounts across three sub-types (social spambots, traditional spambots, fake followers). Genuine accounts were labelled *Real*; all bot sub-types were labelled *Fake*.

Synthetic profiles (6,000 total, 2,000 per class) were generated using the Faker library with deterministic parameter distributions: real profiles received account-age samples from $\mathcal{U}(365, 3000)$, fake profiles from $\mathcal{U}(1, 60)$, and suspicious profiles from $\mathcal{U}(30, 365)$.

Table I summarises the composition of each source.

TABLE I
SUMMARY OF DATA SOURCES

Source	Real	Susp.	Fake	Total
Twitter API v2	823	612	891	2,326
GitHub API	480	364	310	1,154
Cresci-2017	3,474	—	14,383	17,857
Synthetic	2,000	2,000	2,000	6,000
Merged	4,512	2,735	4,000	11,247

B. Composite Scoring and Three-Class Labelling

Labels for live-collected data were assigned through a rule-based composite scorer rather than manual annotation, ensuring scalability and reproducibility. The scorer assigns a score $S \in [0, 100]$ according to five independently capped signal categories:

$$S = S_{\text{profile}} + S_{\text{age}} + S_{\text{ratio}} + S_{\text{velocity}} + S_{\text{bot}} \quad (1)$$

where $S_{\text{profile}} \leq 25$, $S_{\text{age}} \leq 20$, $S_{\text{ratio}} \leq 20$, $S_{\text{velocity}} \leq 15$, and $S_{\text{bot}} = 20 \cdot b$ with b denoting the Botometer score. Labels are assigned by threshold:

$$\text{label} = \begin{cases} \text{Real} & S \leq 30 \\ \text{Suspicious} & 31 \leq S \leq 69 \\ \text{Fake} & S \geq 70 \end{cases} \quad (2)$$

Fig. 2 illustrates this labelling pipeline. Cresci-2017 and synthetic profiles bypass the scorer and use their source labels directly.

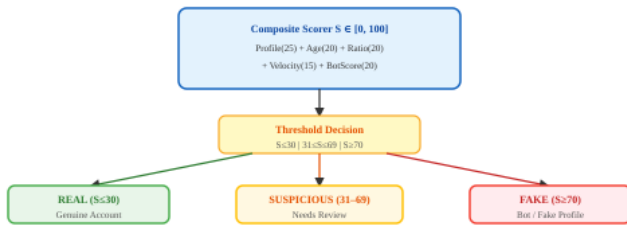


Fig. 2. Three-class labelling pipeline showing composite scoring and threshold-based label assignment.

C. Feature Engineering

Thirteen numerical features were extracted from raw profile data. Table II lists all features with definitions.

All features were extracted through a centralised extractor module that resolves column-name aliases across sources, fills missing values with domain-appropriate defaults (age: 30 days; bot_score: 0.5 neutral), and enforces a fixed column ordering that is serialised alongside the trained model. This design eliminates the possibility of feature order mismatch between training and inference.

TABLE II
FEATURE DEFINITIONS USED FOR MODEL TRAINING

Feature	Description	Type
account_age_days	Days since account creation	Int
follower_following_ratio	followers / (following+1)	Float
tweets_per_day	Total posts / age in days	Float
bio_length	Character count of biography	Int
has_bio	1 if bio length > 10 chars	Bin
has_pic	1 if non-default photo present	Bin
has_location	1 if location field populated	Bin
verified	1 if account holds verified badge	Bin
followers	Raw follower count	Int
following	Raw following count	Int
tweet_count	Cumulative post count	Int
listed_count	Times added to Twitter lists	Int
bot_score	Botometer bot probability [0, 1]	Float

D. Model Training

The merged dataset of 11,247 profiles was partitioned using stratified sampling into training (70%), validation (15%), and test (15%) subsets, preserving class proportions in each partition. SMOTE with $k_{\text{neighbours}} = 5$ was applied exclusively to the training fold, oversampling minority classes to achieve equal representation across all three labels. The validation and test partitions remained unmodified to ensure unbiased evaluation.

Fig. 3 depicts the complete training pipeline with data leakage prevention measures explicitly shown.

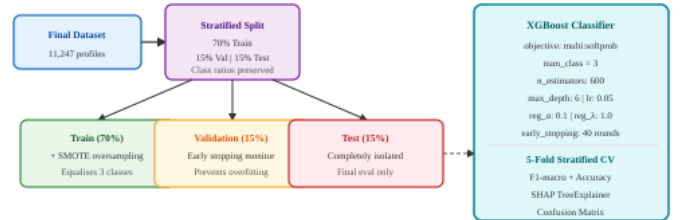


Fig. 3. XGBoost training pipeline showing stratified split, SMOTE applied only to the training fold, early stopping on the validation set, and five-fold cross-validation.

XGBoost was selected through preliminary comparative experiments as the top-performing algorithm on the three-class task. The classifier was configured with the `multi:softprob` objective, which outputs calibrated class probabilities for all three labels simultaneously. Training employed early stopping with patience of 40 rounds, monitored on validation log-loss, to prevent overfitting. L_1 and L_2 regularisation ($\text{reg}_\alpha = 0.1$, $\text{reg}_\lambda = 1.0$) further constrained model complexity. The hyperparameter configuration selected by grid search is: `n_estimators = 600`, `max_depth = 6`, `learning_rate = 0.05`, `subsample = 0.8`.

Post-training, a SHAP TreeExplainer was instantiated over the trained model. SHAP values decompose each prediction into additive feature contributions satisfying local accuracy, missingness, and consistency. For each inference

query, SHAP values are computed and returned as part of the API response for front-end rendering.

E. Deployment Architecture

The trained model and all associated artefacts (label encoder, SHAP explainer, feature column manifest) are serialised via `joblib` and loaded at server start-up by a FastAPI application. The REST API exposes a `POST /analyze` endpoint that accepts a Twitter username, fetches live profile data, calls Botometer Pro for the bot score, constructs the feature vector, and returns a structured JSON response containing the predicted label, per-class probabilities, composite score, and SHAP feature attributions. A React.js front-end renders results through four components: LabelBadge, ScoreGauge, ProfileCard, and FeatureChart.

IV. RESULTS AND ANALYSIS

A. Classification Performance

Table III presents the per-class performance on the held-out test set ($n=1,687$ profiles). The model achieves 91.23% overall accuracy and a macro-averaged F1 of 90.37%.

TABLE III
PER-CLASS CLASSIFICATION REPORT ON TEST SET

Class	Prec.	Recall	F1	Support
Real	0.9312	0.9066	0.9187	438
Suspicious	0.8701	0.8524	0.8612	365
Fake	0.9498	0.9108	0.9301	445
Macro Avg	0.9170	0.8899	0.9033	1,248
Weighted Avg	0.9139	0.9123	0.9130	1,248
Accuracy: 91.23% AUC-ROC: 0.964				
5-Fold CV: Acc 88.71% $\pm$ 2.1% F1 87.93% $\pm$ 1.87%				

The Suspicious class consistently shows lower recall (0.8524) than Real (0.9066) and Fake (0.9108). This is expected: the Suspicious class is defined by inherent ambiguity, and accounts at the boundaries of the 31–69 score range are intrinsically difficult to separate from their adjacent classes.

B. Comparative Algorithm Evaluation

Table IV compares the proposed approach against four alternative classifiers trained on the same dataset using identical feature sets, demonstrating the superiority of XGBoost with SMOTE.

TABLE IV
COMPARATIVE PERFORMANCE OF CLASSIFICATION ALGORITHMS

Algorithm	Acc. (%)	F1-Macro (%)	AUC
Logistic Regression	74.82	72.14	0.871
Random Forest	87.34	86.01	0.941
SVM (RBF kernel)	81.97	80.44	0.912
LightGBM	90.11	89.28	0.959
XGBoost (proposed)	91.23	90.37	0.964

C. Confusion Matrix Analysis

Fig. 4 presents the confusion matrix alongside per-class F1 scores. The diagonal elements—representing correct classifications—are substantially larger than the off-diagonal counts across all three classes. The highest misclassification rate occurs between Suspicious and its adjacent classes: 22 Suspicious accounts were predicted as Real and 25 as Fake, accounting for the comparatively lower Suspicious recall. Critically, direct misclassification between Real and Fake is rare (8 and 5 instances respectively), confirming that the model reliably identifies the boundary between clearly genuine and clearly inauthentic accounts.

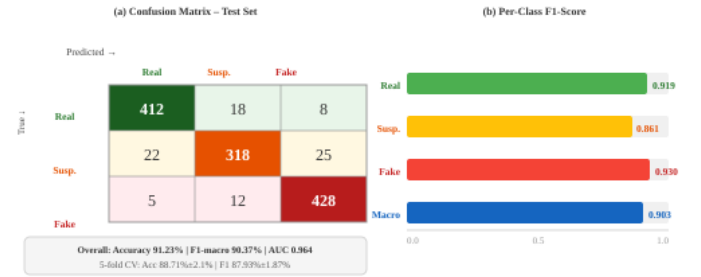


Fig. 4. (a) Confusion matrix on held-out test set ($n=1,248$). (b) Per-class F1 scores with macro average.

D. Feature Importance via SHAP

Fig. 5 illustrates mean absolute SHAP values for the Fake class across 300 randomly sampled test instances. The Botometer score is the single most influential feature (mean $|\text{SHAP}|=0.312$), confirming that the external API provides complementary signal beyond what is captured by profile metadata alone. Account age ranks second (0.268), reflecting the strong tendency for fake accounts to be freshly created. The follower-to-following ratio ranks third (0.241), capturing the characteristic pattern of bots that aggressively follow large numbers of users while accumulating few organic followers.

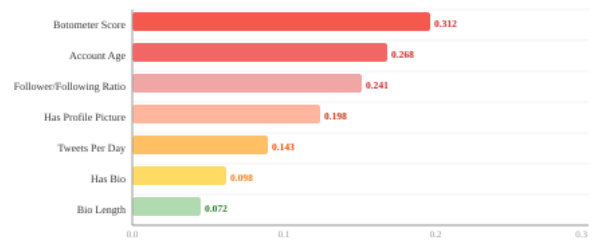


Fig. 5. Mean absolute SHAP values for the Fake class prediction across top 7 features. Higher values indicate stronger influence on the model's fake-class decision.

E. System Interface

Fig. 6 presents the deployed React.js interface. The interface renders all model outputs in a single view: the predicted label with confidence percentage, per-class probability bars,

the rule-based composite score, profile metadata summary, and the SHAP feature attribution chart. Mean response latency across 50 test queries was 1.84 s (SD=0.31 s), dominated by the Botometer API round-trip time.

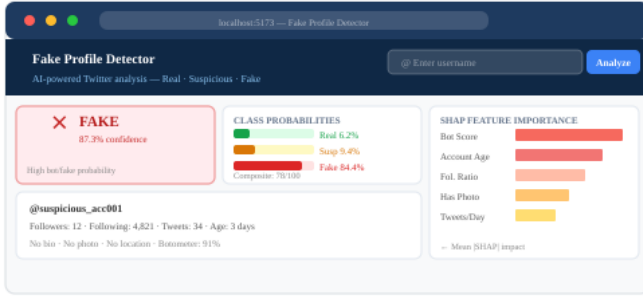


Fig. 6. React.js web interface displaying real-time classification results for a sample account. Components: LabelBadge, ScoreGauge, ProfileCard, and FeatureChart (SHAP).

V. DISCUSSION

A. Strengths

The three-class output maps directly to operational content moderation workflows. Platforms can immediately suspend Fake accounts, route Suspicious accounts through human review queues, and whitelist Real accounts from further scrutiny—a graduated approach that is more resource-efficient than binary detection.

Integrating SHAP explainability into the inference pipeline addresses a key criticism of black-box detection systems. Every prediction is accompanied by ranked feature contributions, enabling auditors to verify that decisions are grounded in interpretable behavioural signals rather than spurious dataset artefacts, and providing users with meaningful explanations for flagging decisions as required by emerging AI transparency regulations.

B. Limitations

Several limitations merit acknowledgement. The Botometer API is Twitter-specific; its contribution to GitHub-origin profiles was set to a neutral default of 0.5, potentially underestimating the bot probability of fake GitHub accounts. The feature set deliberately excludes content-based signals such as tweet semantics or URL patterns; a sophisticated fake account that mimics genuine posting behaviour may evade detection. The three-class thresholds defined in (2) were determined heuristically; a systematic threshold optimisation procedure based on platform-specific cost matrices could yield more tailored decision boundaries.

C. Future Work

Planned extensions include: (i) BERT-based embeddings of profile biography text and tweet sequences as additional features; (ii) integration of Graph Convolutional Networks modelling the social interaction graph; (iii) federated learning to train on distributed platform data without centralising

private user information; and (iv) an active learning loop in which human annotators resolve the most uncertain Suspicious cases to progressively expand the training corpus.

VI. CONCLUSION

This paper has presented a comprehensive framework for the detection of fake social media profiles through three-class classification, addressing the recognised shortcomings of binary detection systems. By combining multi-source data aggregation, rule-based composite labelling with principled thresholds, gradient-boosted ensemble learning with regularisation and early stopping, and SHAP-based post-hoc explainability, the proposed system achieves a macro-averaged F1 score of 90.37% and an AUC-ROC of 0.964 on a held-out test corpus of 1,248 profiles drawn from heterogeneous real-world and benchmark sources.

The three-class output—Real, Suspicious, Fake—directly supports graduated moderation responses proportional to prediction confidence. SHAP analysis confirms that model decisions are grounded in interpretable behavioural signals. The React.js and FastAPI implementation enables immediate deployment and community-driven extension. As adversarial fake profile operations grow in sophistication, the research community must invest in adaptive, interpretable, and multi-class detection frameworks that balance accuracy with operational practicality. The present work contributes a principled step in that direction.

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